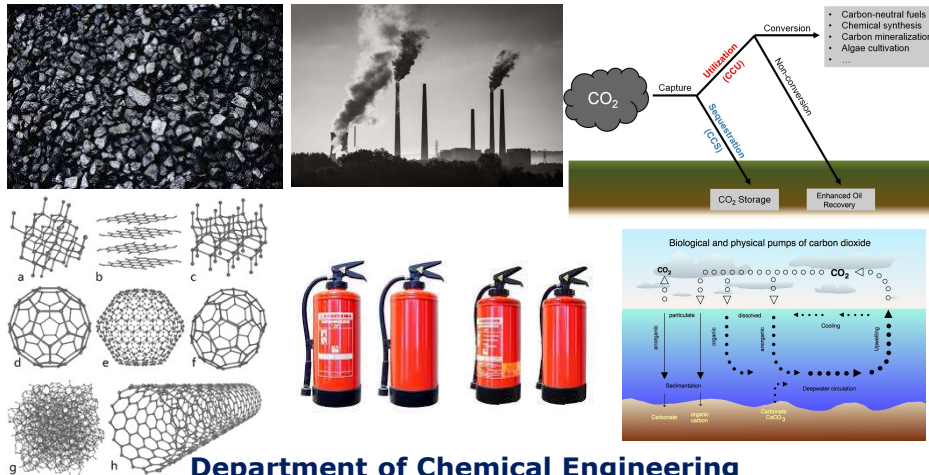


CH447 – CO₂ Capture, Sequestration and Utilization



Introduction



Department of Chemical Engineering

National Institute of Technology, Warangal

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1

Biosphere



- A region of the surface and atmosphere of the earth occupied by living organisms of approximate of 1.75 million species.
- This covers a 20 km from the bottom of the ocean to the highest point in the atmosphere.
- Essential requirements – light, heat, water, food and habitats
- Biotic (living)
 - Autotrophic (producers) – green plants and algae, phytoplankton
 - Heterotrophic (consumers) – animal life ex: fish, mammals, insects, birds etc.
 - Decomposers – bacteria and fungi that produces chemical components from waste materials
- Abiotic (non-living) components
 - Basic elements – Phosphorous and Nitrogen
 - Compounds – Water, oxygen, CO₂ etc.

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2

Biosphere



- Dynamic equilibrium is maintained in a healthy system by providing sufficiently small amounts of nutrients(C, N, P).
 - Chemical elements circulate between these organisms and environment through natural cycles.
 - Hydrologic cycle
 - Biogeochemical cycles
 - Carbon
 - Nitrogen
 - Phosphorous
 - Sulphur
- Nutrient Cycles**
- Feedback loop is maintained by which dynamic equilibrium exists in the environment.
 - Large fluctuations in these balances will destroy environment.

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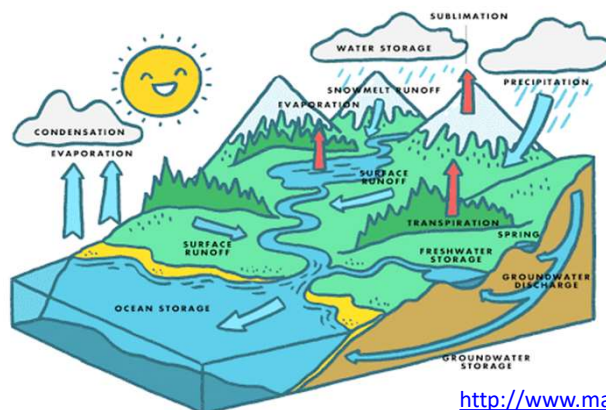
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Hydrologic Cycle



- More than 97% of water is in oceans and 2% is locked in poles and glaciers.
- Remaining 1% constitutes rivers, lakes and ponds – total supply to human needs. This also contains reusable water.



<http://www.matfaint.com/grc-hydro.html>

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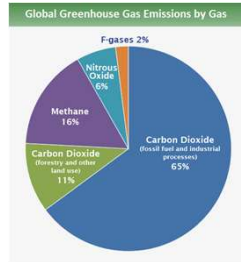
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Nutrient Cycles – Carbon Cycle



- Carbon cycle involves formation, transformation and decomposition.
- Producers by photosynthesis reduces CO_2 to organic carbon which is consumed by consumers and decomposers.
- Human activity releases roughly 7 billion tonnes of carbon per year.



Source	Carbon Reservoir (Billion Tonnes)
Oceans	40,000
Rocks and minerals	5,000 to 10,000
Atmosphere	750

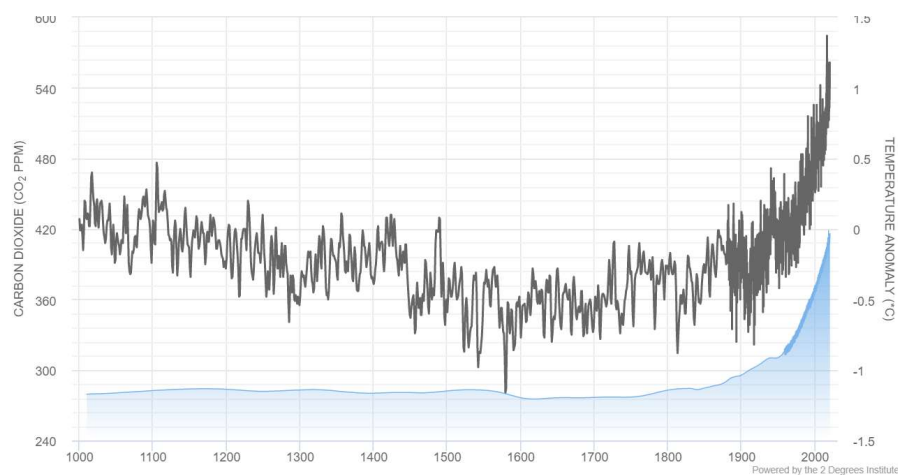
- Additional process involves the non-biological process of combustion.

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Nutrient Cycles – Carbon Cycle


www.co2.earth

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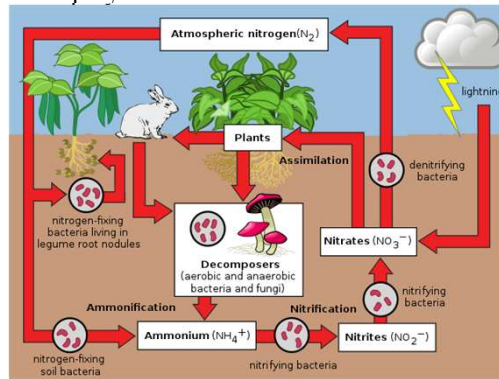
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Nutrient Cycles – Nitrogen Cycle



- Nitrogen constitutes ~79% atmosphere gaseous phase.
- It must be fixed before it can be utilized by plants and animals by biological and physical means.



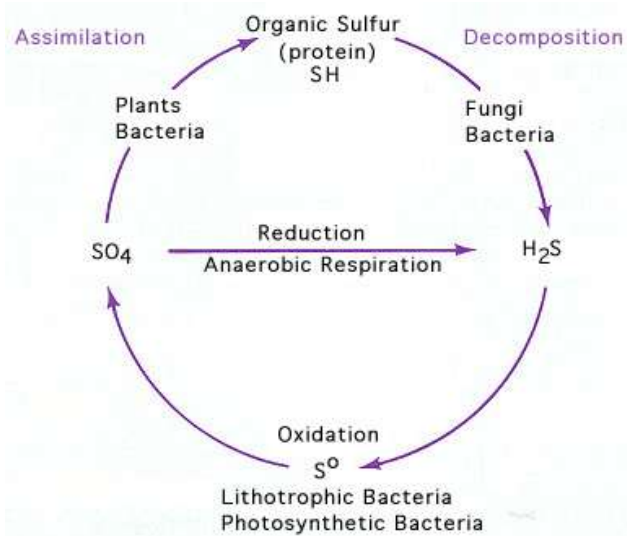
- Industrially fixing of nitrogen – **Urea, combustion of fuels.**
- Eutrophication – Excessive growth of algae in sea bed

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Nutrient Cycles – Sulphur Cycle



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Nutrient Cycles – Sulphur Cycle



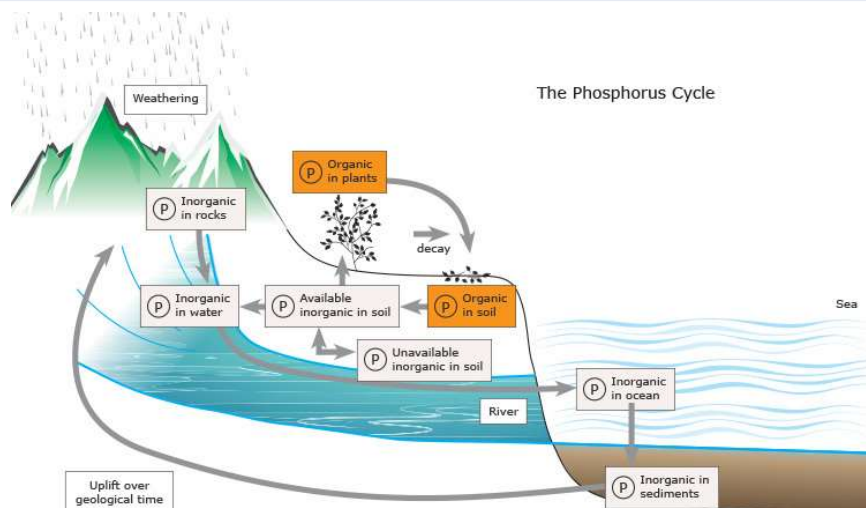
- Sulphur is a basic constituent in plants and animals.
- Sulphur dioxide (SO_2), Hydrogen Sulphide (H_2S) and Sulphate ion (SO_4^{2-}) are common forms of Sulphur source.
- Sulphate ion is reduced to $-\text{SH}$ group by plants and bacteria.
- *Desulfovibrio* bacteria – Anerobic reduction to H_2S at ocean bottom.
- *Thiobacillus* bacteria– Aerobic oxidization of H_2S to elemental Sulphur
- Sulphur in atmosphere is received by bacterial emissions, fuel burning, volcanic emissions.
- Most of the SO_2 and H_2S converted to SO_3 which then comes by precipitation.
- Higher Sulphur content causes **acid rains** which effects plants.

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Nutrient Cycles – Phosphorous Cycle



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Nutrient Cycles – Phosphorous Cycle



- Phosphorous is important in living tissue growth which plays role in energy transfer of metabolic process.
- It cannot return into atmosphere as it is not a gas at normal temperatures.
- Rock and natural phosphate deposits are sources.
- Plants assimilate inorganic phosphate from soil and convert them into adenosine di and tri phosphates.
- *Phytoplankton* – converts inorganic to organophosphates which is further consumed by zooplanktons and organisms.
- When organisms die, phosphates are released into water by bacterial decomposition.
- Most of the phosphorous is moving form land to sea beds.
- Lack of phosphorous causes soil infertility which effects natural food chain.

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Population Data and Density



<https://ourworldindata.org/world-population-cartogram>

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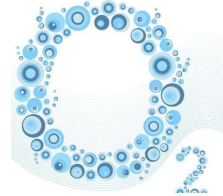
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Primary Resources Needed



- ✓Water
- ✓Food
- ✓Air
- ✓Energy
- ✓Healthcare
- ✓Buildings



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Countries Classification



IEO regional definitions – OECD Regions

OECD Americas = United States, Canada, Chile, and Mexico.

OECD Europe = Austria, Belgium, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, and United Kingdom. (Note: Israel is included in OECD Europe for statistical reporting purposes. Latvia became an OECD member country on July 16, 2016, but it is not reported in OECD Europe for IEO2018.)

OECD Asia = Australia, Japan, New Zealand, and South Korea.

IEO regional definitions – non-OECD Regions

Non-OECD Europe and Eurasia = Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Faroe Islands, Georgia, Gibraltar, Kazakhstan, Kosovo, Latvia, Lithuania, Macedonia, Malta, Moldova, Montenegro, Romania, Russia, Serbia, Tajikistan, Turkmenistan, Ukraine, and Uzbekistan. (Note: Latvia became an OECD member country on July 16, 2016, but it is reported in non-OECD Europe and Eurasia for IEO2018.)

Source: EIA, International Energy Outlook 2018

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Countries Classification



Non-OECD Asia = Afghanistan, American Samoa, Bangladesh, Bhutan, Brunei, Burma (Myanmar), Cambodia (Kampuchea), China, Cook Islands, Fiji, French Polynesia, Guam, Hawaiian Trade Zone, Hong Kong, India, Indonesia, Kiribati, Laos, Macau, Malaysia, Maldives, Mongolia, Nauru, Nepal, New Caledonia, Niue, North Korea, Pakistan, Papua New Guinea, Philippines, Samoa, Singapore, Solomon Islands, Sri Lanka, Taiwan, Thailand, Timor-Leste (East Timor), Tonga, U.S. Pacific Islands, Vanuatu, Vietnam, and Wake Islands.

Middle East = Bahrain, Iran, Iraq, Jordan, Kuwait, Lebanon, Oman, Palestinian Territories, Qatar, Saudi Arabia, Syria, United Arab Emirates, and Yemen.

Africa = Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Cape Verde, Central African Republic, Chad, Comoros, Congo (Brazzaville), Congo (Kinshasa), Côte d'Ivoire, Djibouti, Egypt, Equatorial Guinea, Eritrea, Ethiopia, Gabon, The Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Libya, Madagascar, Malawi, Mali, Mauritania, Mauritius, Morocco, Mozambique, Namibia, Niger, Nigeria, Reunion, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Somalia, South Africa, South Sudan, St. Helena, Sudan, Swaziland, Tanzania, Togo, Tunisia, Uganda, Western Sahara, Zambia, and Zimbabwe

Non-OECD Americas = Antarctica, Antigua and Barbuda, Argentina, Aruba, The Bahamas, Barbados, Belize, Bermuda, Bolivia, Brazil, British Virgin Islands, Cayman Islands, Colombia, Costa Rica, Cuba, Dominica, Dominican Republic, Ecuador, El Salvador, Falkland Islands, French Guiana, Greenland, Grenada, Guadeloupe, Guatemala, Guyana, Haiti, Honduras, Jamaica, Martinique, Montserrat, Netherlands Antilles, Nicaragua, Panama, Paraguay, Peru, Puerto Rico, St. Kitts and Nevis, St. Lucia, St. Pierre and Miquelon, St. Vincent/Grenadines, Suriname, Trinidad and Tobago, Turks and Caicos Islands, Uruguay, United States Virgin Islands, and Venezuela.

Source: EIA, International Energy Outlook 2018

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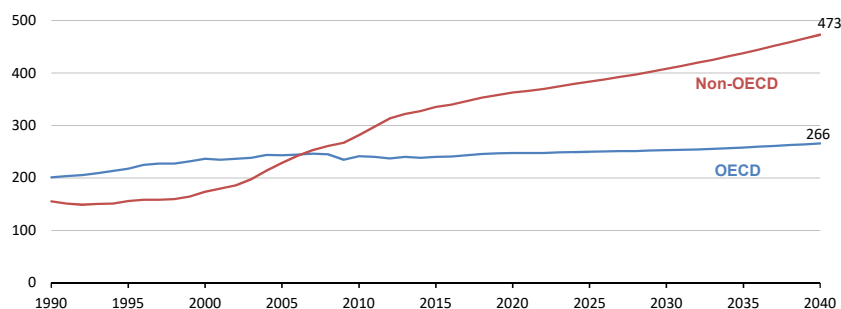
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World Energy Consumption



quadrillion Btu



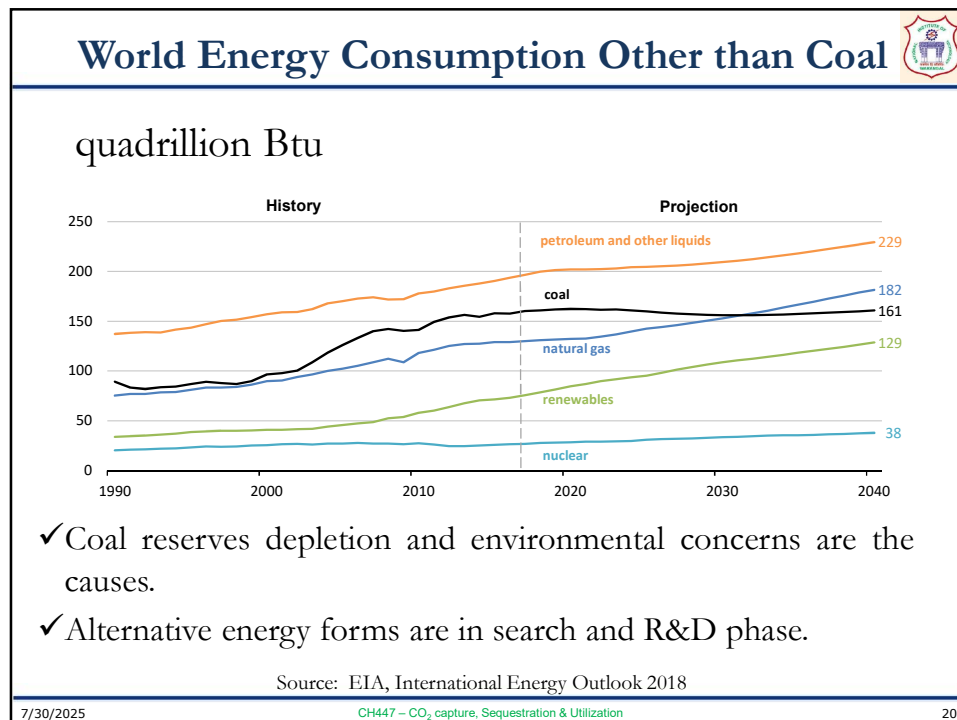
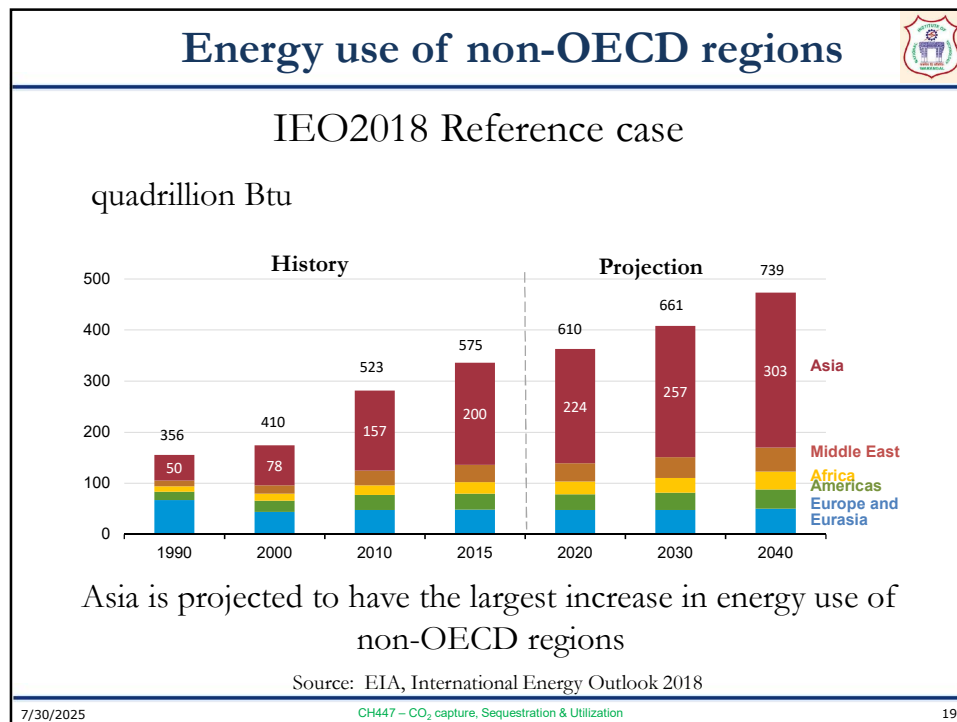
Non-OECD nations are projected to account for 64% of the 739 quadrillion Btu global energy consumption by 2040

Source: EIA, International Energy Outlook 2018

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OECD GHG Emissions and INDIA

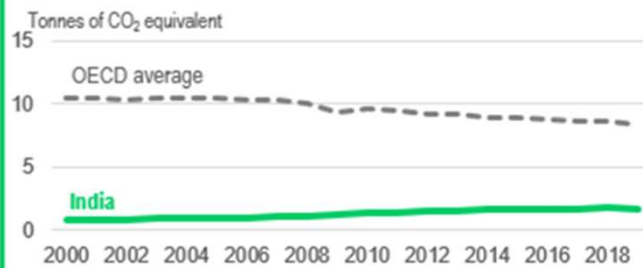


Environment

More than 3/4 of the population is exposed to harmful levels of air pollution.

Carbon dioxide emissions have been stable in recent years.

Greenhouse gas emissions per capita



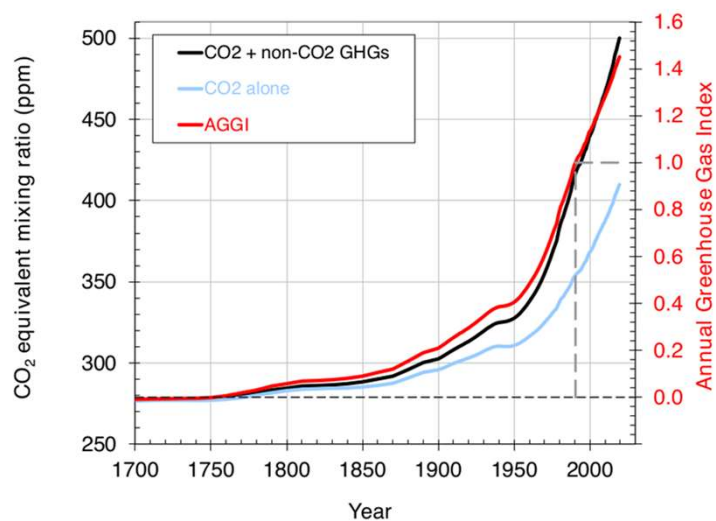
<https://www.oecd.org/economy/india-economic-snapshot/>

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Annual GHG Index



<https://www.co2.earth/annual-ghg-index-aggi>

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Energy Demands



- Energy consumption in the non-OECD countries began to exceed OECD consumption in 2007 and is projected to reach nearly two-thirds of the 739 quadrillion Btu global energy consumption in 2040
- The world's energy consumption through 2040 increases, on average, for all fuels in the IEO2018 Reference case
- The IEO2018 side cases show higher economic growth drives increasing energy consumption, while services or manufacturing pathways to growth modulate that consumption
- **Per capita energy consumption** in India and Africa remain comparatively low despite high economic growth in the IEO2018 side cases
- IEO2018 side cases highlight the need to further explore the relationship between high economic growth, relative sizes of the services and manufacturing sectors, and energy consumption

Source: EIA, International Energy Outlook 2018

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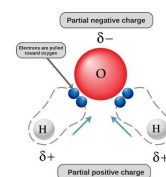
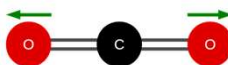
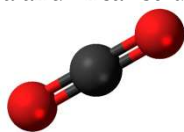
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What is CO₂



- Carbon dioxide (CO₂) is a triatomic molecule with a molecular weight of 44 Da and linear structure.



- It is a colour less gas at room temperature and pressure with a sublimation temperature (-78 °C).
- Inert gas which is not an explosive, not flammable which does not support combustion. It is widely used in fire extinguishers and suppression systems.
- Natural source of CO₂ is volcanic eruptions, forest fires, plants and animal respirations.
- Basic constituent of photosynthesis which is essential for green plants growth.

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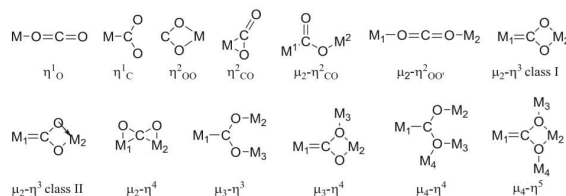
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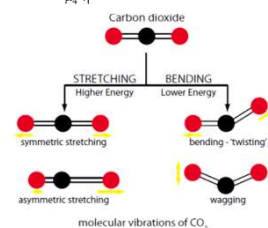
What is CO₂



- The chemical reactivity of CO₂ is determined by the polarization of carbon oxygen bonds.
- Coordination of CO₂ with metal can significantly alter the chemistry of molecule by altering the electron distribution and molecular geometry.



- CO₂ has three vibrational modes (symmetric, asymmetric stretching and bending).
- Asymmetric stretch molecule is infrared active which causes its ability to greenhouse gas.

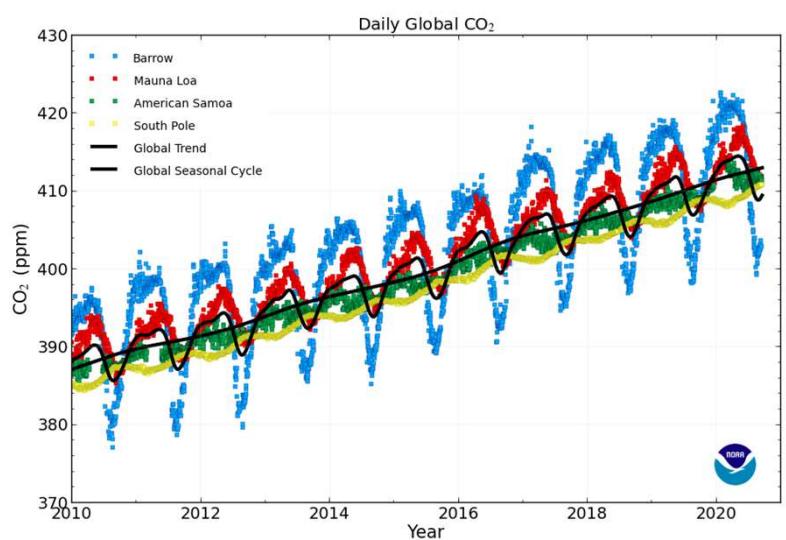


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CO₂ Recent Global Concentrations



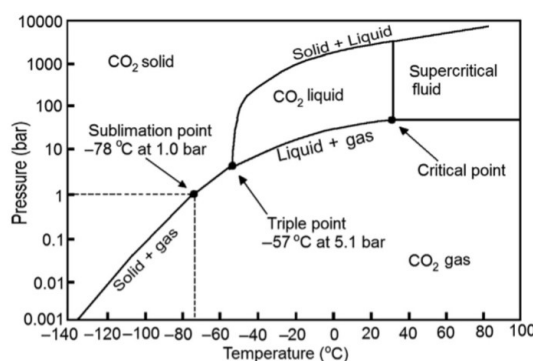
https://www.esrl.noaa.gov/gmd/ccgg/trends/gl_trend.html

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CO₂ Phase Behaviour



- CO₂ Supercritical region is distributed over a range of temperatures and pressures. This property made its applicability of usage in green solvents, chromatography, extractions.
- Decaffeination of coffee beans is one of the largest commercial use.

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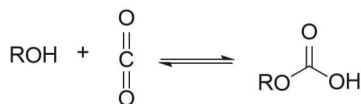
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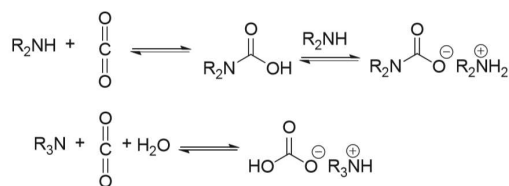
CO₂ Reactions with common reagents



- CO₂ can react with water and alcohols to form carbonic acid which can be used to alter the miscibility of the solvents.
- The carbonic acid can act as a acid catalyst in reactions which will be neutralized by itself when the pressure is released at the end of the reaction.



- CO₂ can react with amines to form Carbamic acids and ammonium carbamate salts.



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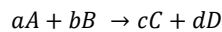
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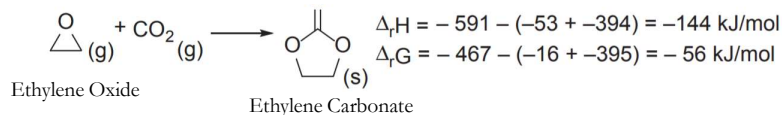
CO₂ Reactions Kinetics and Thermodynamics



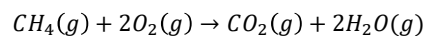
- CO₂ is thermodynamically stable with heat of formation of -394 kJ/mol. The reactions can also be exothermic if CO₂ is involved.



$$\Delta H_{rxn}^{\circ} = \underbrace{\left[c\Delta H_f^{\circ}(C) + d\Delta H_f^{\circ}(D) \right]}_{\text{products}} - \underbrace{\left[a\Delta H_f^{\circ}(A) + b\Delta H_f^{\circ}(B) \right]}_{\text{reactants}}$$



- Calculate the standard heat of reaction for the following reaction. Given the heat of formation for CH₄, CO₂ and H₂O are -75, -394 and -284 kJ/mol respectively



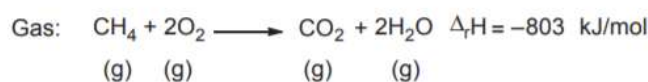
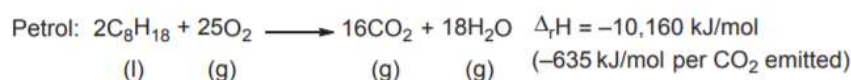
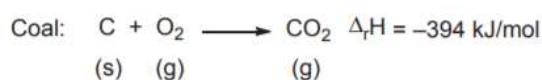
$$\Delta H_r^{\circ} = [1(-394) + 2(-284)] - [1(-75) + 2(0)] = -887 \text{ kJ/mol}$$

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Thermodynamics Properties



- The energy produced by the gaseous fuels is more for every mole of CO₂ produced.
- The ratio H/C in fuel also makes a difference in energy produced with less amount of CO₂ produced.
- Free energy of reaction plays important role in fixing the equilibrium for a reaction.

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Thermodynamics Properties

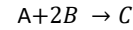


Table 1.1 Thermodynamic Parameters for Chemicals Relevant to Carbon Dioxide Chemistry^{2,40}

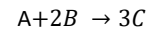
Chemical (State)	$\Delta_f H^\circ$ (kJ/mol)	$\Delta_f G^\circ$ (kJ/mol)
H ₂ O (g, l)	-242 (g), -286 (l)	-229 (g), -241 (l)
CO ₂ (g)	-394	-395
CO (g)	-111	-135
HCO ₂ H (g, l)	-379 (g), -425 (l)	-354 (g), -368 (l)
H ₂ CO (g, l)	-116 (g)	-111 (g), -102 (l) ⁴¹
CH ₃ OH (g, l)	-205 (g), -238 (l)	-171 (l)
CH ₃ OCH ₃ (g)	-184.1	-113 ⁴¹
CH ₄ (g)	-75	-1.5
C ₈ H ₁₈ (octane) (g, l)	-209 (g), -250 (l)	0 (g), -13 (l)
H ₂ C=CH ₂ (g)	+52	+68
CH ₃ CH ₂ OH (g, l)	-234 (g), -276 (l)	-169 (g), -181 (l)
HOCH ₂ CH ₂ OH (g, l)	-394 (g), -460 (l)	-313 (g), -339 (l)
Ethylene oxide (g, l)	-53 (g), -96 (l)	-16 (g), -34 (l)
Ethylene carbonate (g, l, s)	-503 (g), -683 (l), -591 (s)	-467 (s)
CH ₃ OCOOCH ₃ (l)	-614 ⁴⁰	-492 ⁴¹
Propylene oxide (g, l)	-95 (g), -123 (l)	-33 (g), -36 (l)
Propylene carbonate (g, l)	-564 (g), -613 (l)	-475 (l)
C ₆ H ₆ (benzene) (g, l)	+83 (g), +49 (l)	+130 (g), +119 (l)
C ₆ H ₅ CO ₂ H (s)	-85	-255
NH ₃ (g)	-46	-19
CO(NH ₂) ₂ (g, s)	-236 (g), -333 (s)	-208 (s)
MgO (s)	-601	-548
CaO (s)	-635	-580
MgCO ₃ (s)	-1112	-1012
CaCO ₃ (s)	-11,004	-1130 ⁴¹
K ₂ CO ₃ (s)	-1150	-978

^a $\Delta_f G^\circ$ are reported relative to the values for the constituent elements.

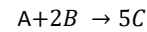
$$\Delta G_r = \Delta H_r - T\Delta S_r$$



- As temperature increases reaction equilibrium shifts towards reactants.



- As temperature increases reaction equilibrium depends on the heat of reaction and entropy values.



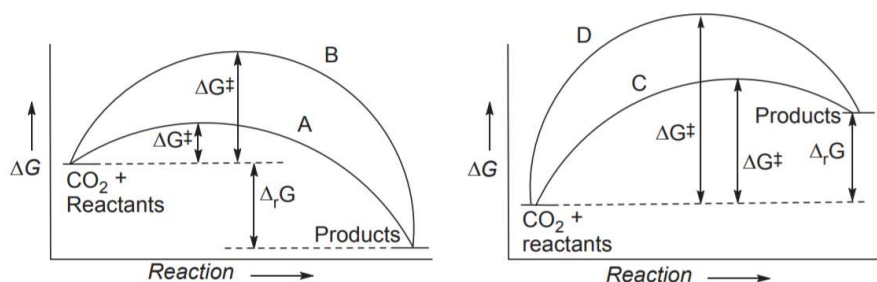
- As temperature increases reaction equilibrium shifts towards products.

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Reaction pathways



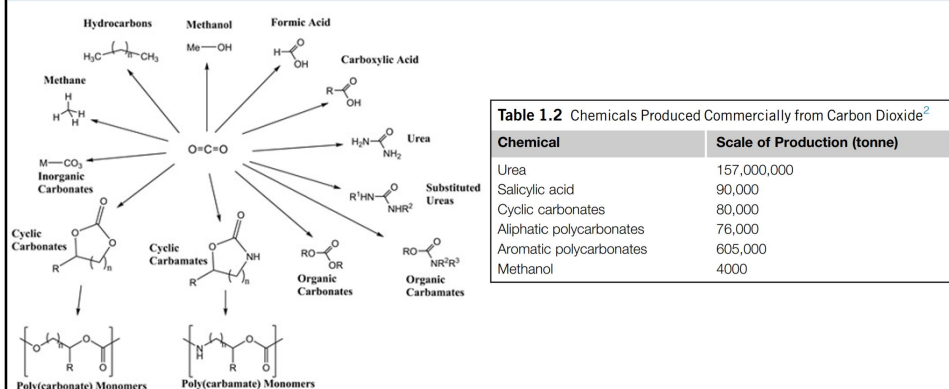
- Reaction schemes A, B have negative ΔG should give more products at equilibrium.
- Reaction schemes C, D have positive ΔG should give less products at equilibrium.
- Catalyst reduces the activation energy barrier to form the products. Catalyst is of no use in case of schema C.
- Removal of one of the product will help progress in schema C.

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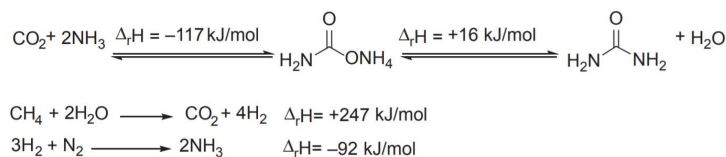
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Organics Chemicals from CO₂



<https://www.chemistryworld.com/green-and-sustainable-chemistry/carbon-dioxide-greenhouse-gas-or-useful-chemical-feedstock/3009130.article>



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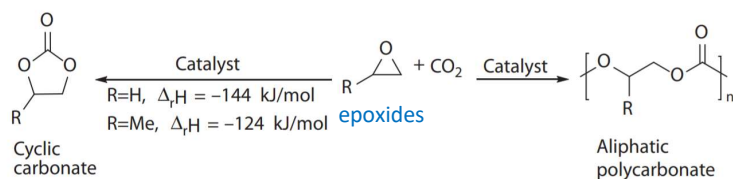
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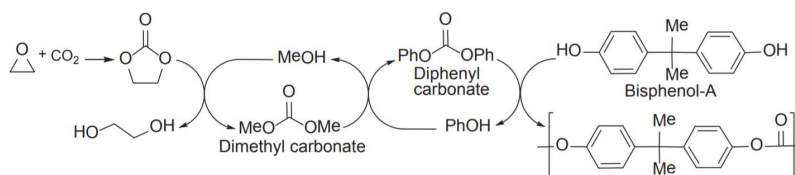
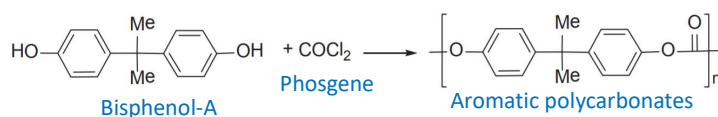
Organics Chemicals from CO₂



Electrolytes in
Lithium ion
batteries



Phosgene and
bisphenol are
poisonous

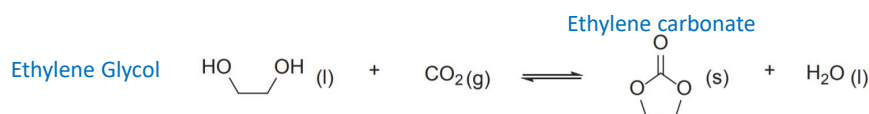


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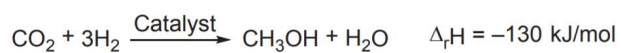
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Organics Chemicals from CO₂



Methanol
formation



Less hydrogen
consuming reaction



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References



1. Peter Styring, Elsje Alessandra Quadrelli, Katy Armstrong, Carbon dioxide utilization: Closing the Carbon Cycle, Elsevier, 2015.
2. Goel M, Sudhakar M, Shahi RV, Carbon Capture, Storage and, Utilization: A Possible Climate Change Solution for Energy Industry, TERI, Energy and Resources Institute, 2015.
3. Amitava Bandyopadhyay, Carbon Capture and Storage, CO₂ Management Technologies, CRC Press, 2014. 4.
4. Fennell P, Anthony B, Calcium and Chemical Looping Technology for Power Generation and Carbon Dioxide (CO₂) Capture, Woodhead Publishing Series in Energy: No. 82, 2015.
5. Mercedes Maroto-Valer M, Developments in Innovation in Carbon Dioxide Capture and Storage Technology: Carbon Dioxide Storage and Utilization, Vol 2, Woodhead Publishing Series in Energy, 2014.

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